

Project 1: Data Cleaning & Preparation

Industrial Training – Data Analytics

Objective

The objective of this project was to transform a raw e-commerce dataset into a clean and reliable dataset suitable for further analysis. The focus was on identifying and resolving common data quality issues such as missing values, duplicate records, inconsistent formatting, and incorrect data types.

Dataset Overview

The dataset contains transactional information related to customer orders placed on an e-commerce platform. It includes customer details, product information, payment methods, shipping details, order status, and transaction values.

Dataset Summary

- Total Records: **1,200**
- Total Columns: **14**

The dataset includes the following fields:

- Order ID
 - Date
 - Customer ID
 - Product
 - Quantity
 - Unit Price
 - Shipping Address
 - Payment Method
 - Order Status
 - Tracking Number
 - Items in Cart
 - Coupon Code
 - Referral Source
 - Total Price
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Data Cleaning Process

1. Missing Value Identification

The dataset was examined to identify blank or null values across all columns.

Action Taken

- Checked every column for missing values.
- Verified that all missing records were handled appropriately.

Result

No missing values remained after the cleaning process.

2. Duplicate Record Verification

Duplicate records can negatively impact analysis and reporting.

Action Taken

- Checked the dataset for duplicate Order IDs.
- Verified that every order was uniquely represented.

Result

No duplicate Order IDs were found in the cleaned dataset.

3. Date Formatting

Consistent date formatting is essential for sorting, filtering, and time-based analysis.

Action Taken

- Standardized all date values into a consistent Excel date format.
- Verified that all entries were recognized as valid dates.

Result

All dates were correctly formatted and ready for analysis.

4. Data Type Validation

Each column was reviewed to ensure that it contained the correct type of data.

Examples include:

- Date stored as Date format
- Quantity stored as Numeric
- Unit Price stored as Numeric
- Total Price stored as Numeric
- IDs and category fields stored as Text

This validation helps avoid calculation errors during analysis.

5. Consistency Check

The cleaned dataset was reviewed to ensure consistency across all records.

Checks included:

- Standardized text values
 - Consistent formatting
 - Proper numerical values
 - Complete records
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Validation Results

Validation Check	Status
Missing Values	Completed
Duplicate Records	No duplicates found
Date Formatting	Standardized
Data Types	Validated
Dataset Ready for Analysis	Yes

Outcome

The dataset was successfully cleaned and prepared for analytical tasks. The final dataset contains consistent, complete, and reliable information that can be used for exploratory data analysis, SQL querying, dashboard creation, and business reporting.

Skills Applied

- Data Cleaning
 - Data Validation
 - Data Preparation
 - Excel Functions
 - Duplicate Detection
 - Missing Value Handling
 - Data Formatting
 - Quality Assurance
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Conclusion

Data cleaning is one of the most important stages of the data analytics process because the quality of insights depends on the quality of the underlying data. During this project, the dataset was carefully reviewed, validated, and cleaned by removing inconsistencies, verifying data types, checking for duplicates, and ensuring that no missing values remained. The resulting dataset is accurate, consistent, and ready for further analysis in subsequent projects.